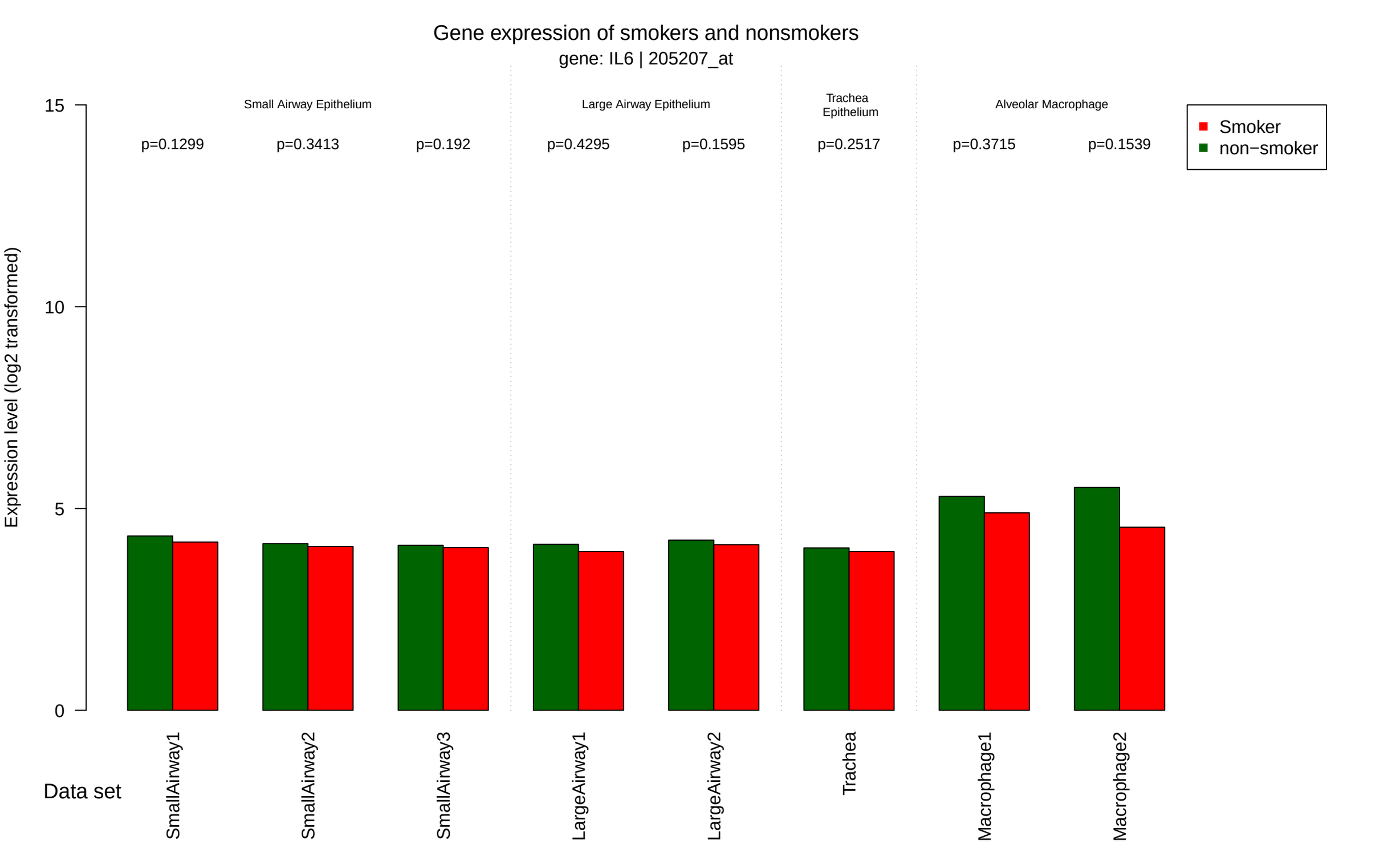
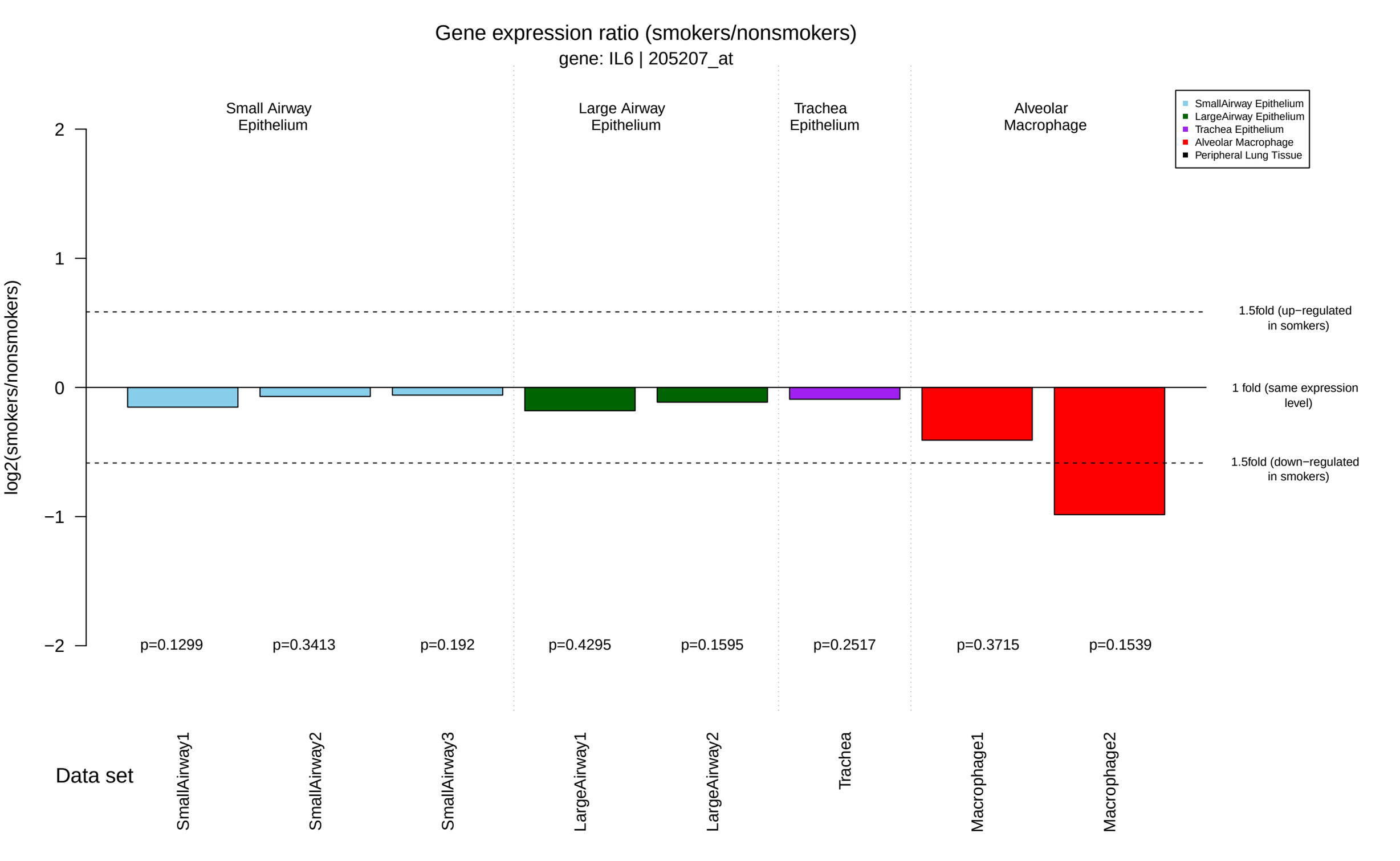
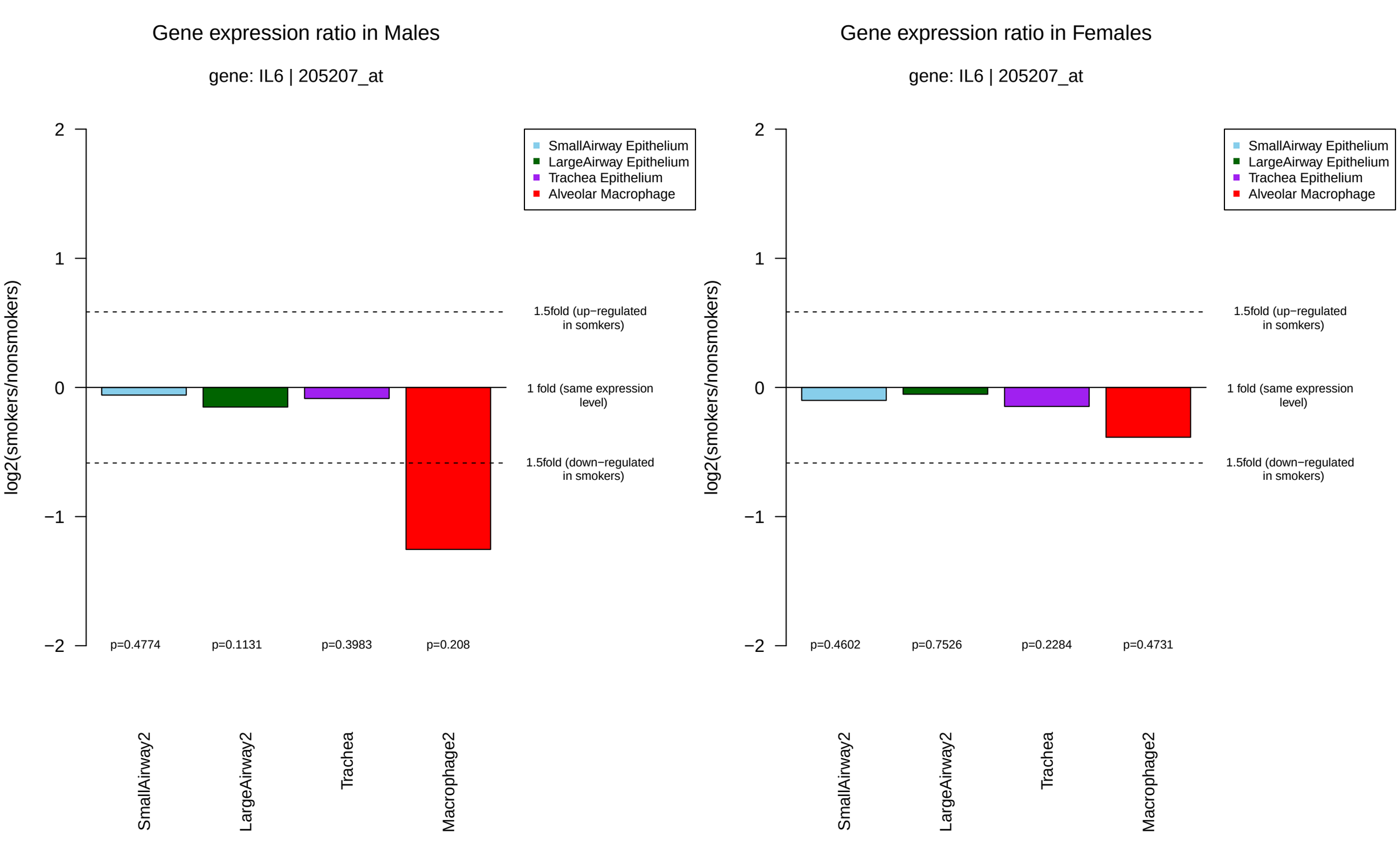
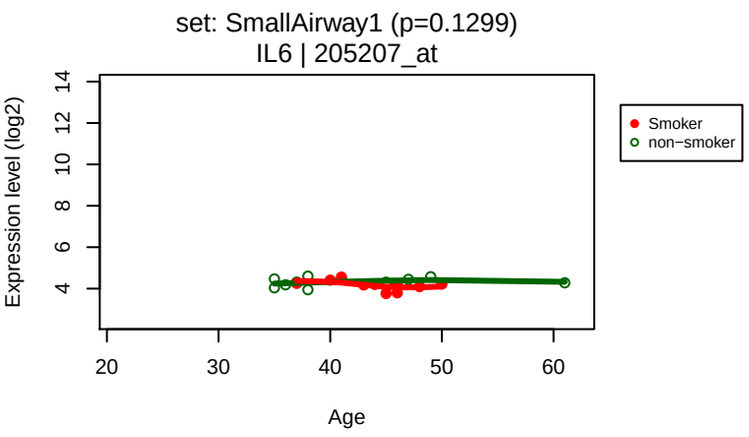
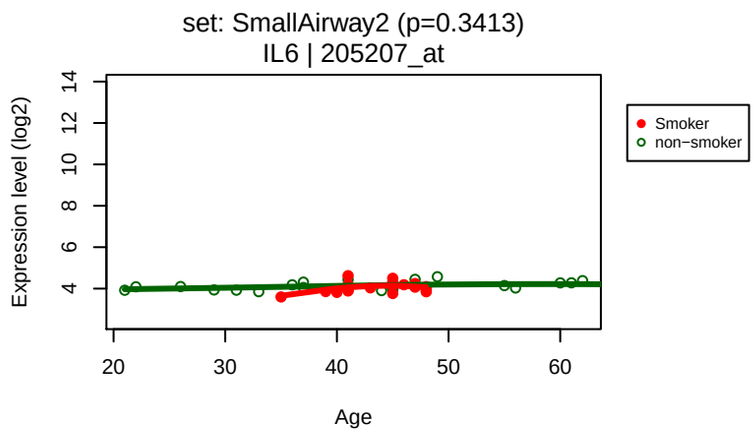
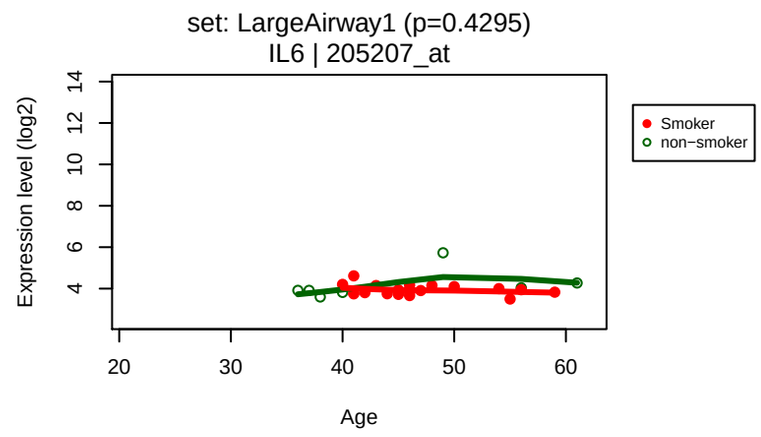

Gene expression of smokers and nonsmokers
gene: IL6 | 205207_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1299
p=0.3413
p=0.192
p=0.4295
p=0.1595
p=0.2517
p=0.3715
p=0.1539
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

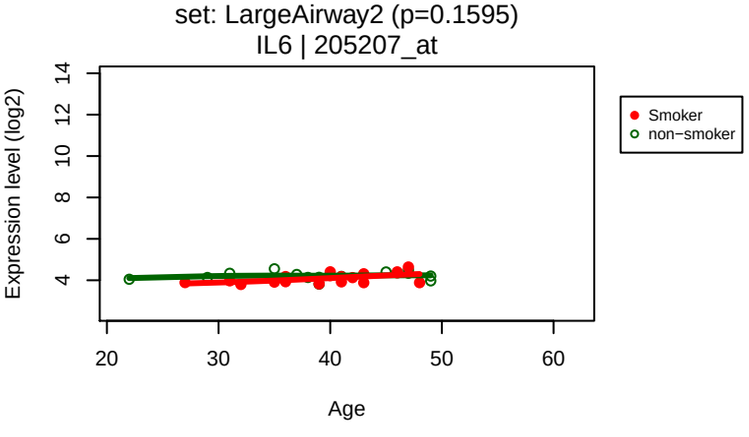

Gene expression ratio (smokers/nonsmokers)
gene: IL6 | 205207_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1299
p=0.3413
p=0.192
p=0.4295
p=0.1595
p=0.2517
p=0.3715
p=0.1539
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

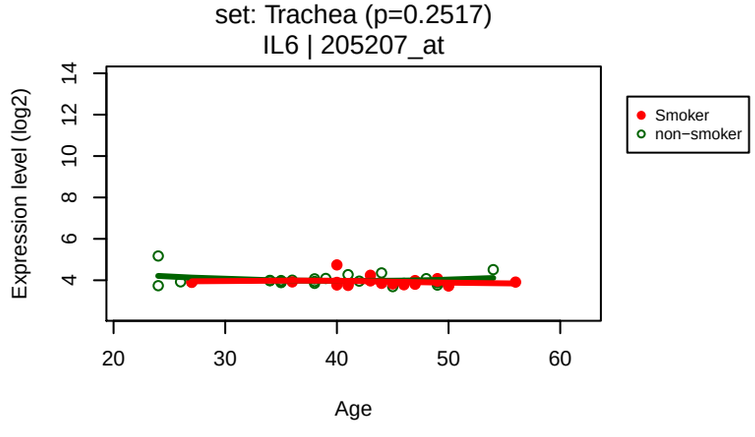

Gene expression ratio in Males
gene: IL6 | 205207_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4774
p=0.1131
p=0.3983
p=0.208
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL6 | 205207_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4602
p=0.7526
p=0.2284
p=0.4731
SmallAirway2
LargeAirway2
Trachea
Macrophage2

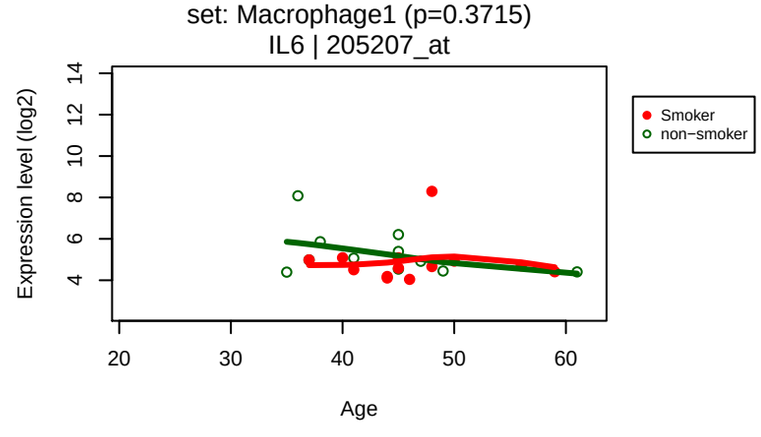

set: SmallAirway1 (p=0.1299)
IL6 | 205207_at
Expression level (log2)
Age
Smoker
non−smoker

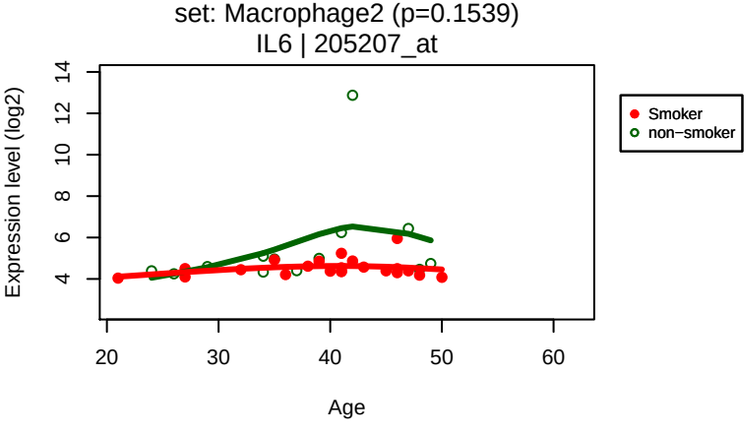

set: SmallAirway2 (p=0.3413)
IL6 | 205207_at
Expression level (log2)
Age
Smoker
non−smoker

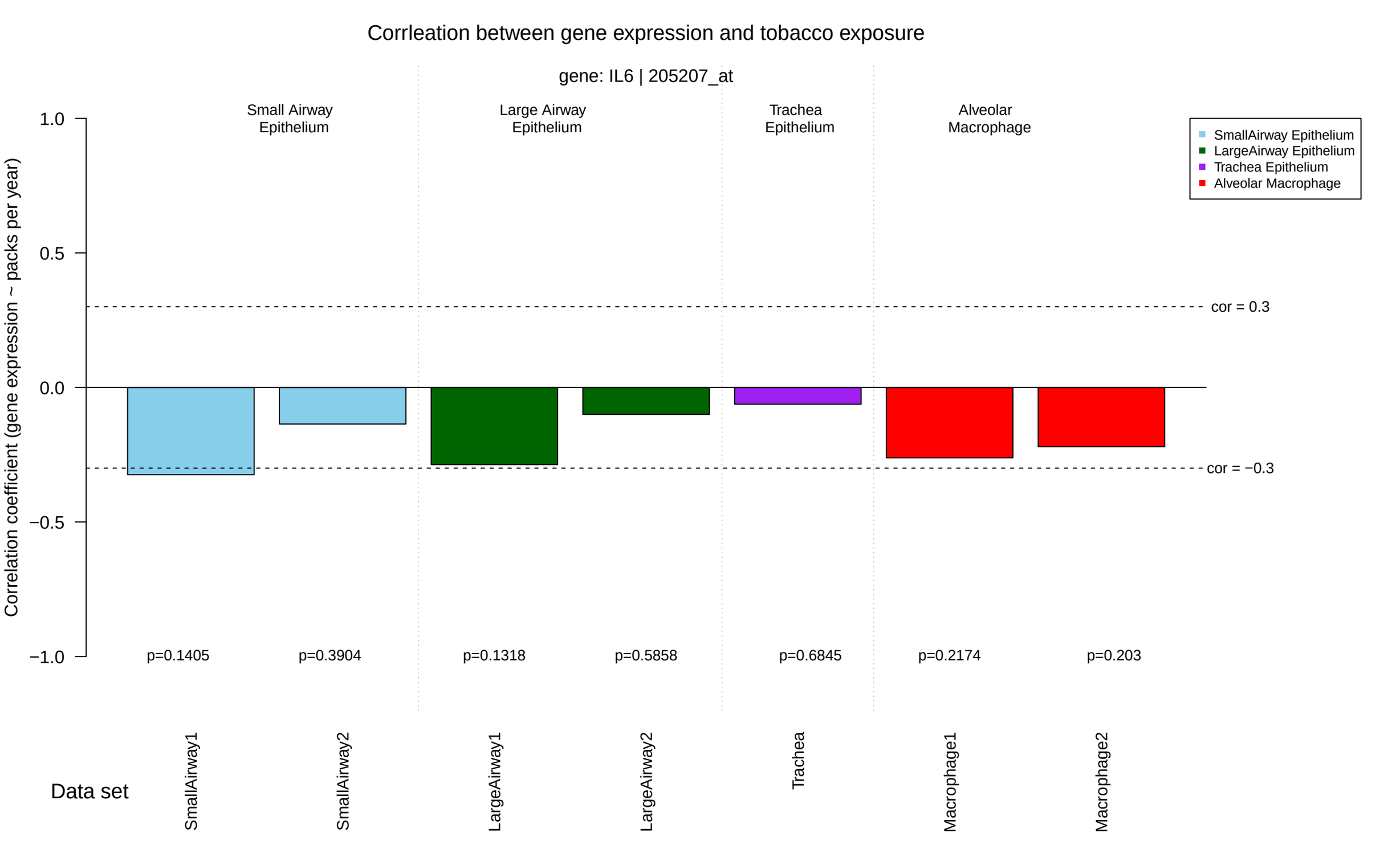

set: LargeAirway1 (p=0.4295)
IL6 | 205207_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1595)
IL6 | 205207_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2517)
IL6 | 205207_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3715)
IL6 | 205207_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1539)
IL6 | 205207_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL6 | 205207_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1405
p=0.3904
p=0.1318
p=0.5858
p=0.6845
p=0.2174
p=0.203
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2